(DEEMED TO BE UNIVERSITY)

TITLE :

SMART DIGITAL LIBRARY BOOK PORTAL

PROJECT BASED LEARNING FROM DSA

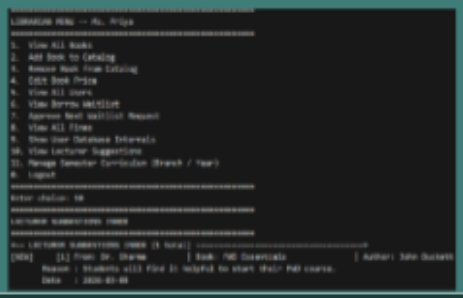
ABSTRACT:
A Library Management System designed to simplify the organization and access of library resources. The platform enables students and lecturers to search and borrow books, track fine dues, and suggest new books or subjects. It also allows librarians to efficiently manage books, users, and library records, improving overall library operations.

CO-WISE IMPLEMENTATION

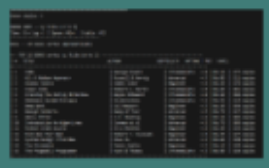
CO1 – Linear search using find() retrieves users and books by ID or email. Password validation scans input characters linearly counting digits and letters.

CO2 – Users, books, issued records, and history stored and managed as arrays. Core operations include insert via push, search via find, and delete via filter.

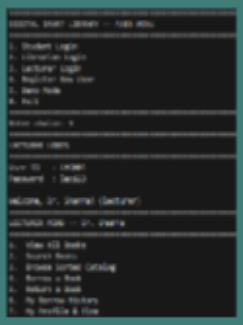
CO 1 AND CO 2



CO3 – Suggestion list and reading history follow FIFO queue insertion behavior. Toast notification system processes one message at a time mimicking a queue.



CO4 – localStorage functions as a key-value hash map managed through the KEYS object. Subject-book relationships stored using composite string keys as hash identifiers.



CO5 – Issued books, fines, suggestions, and history all managed using linear arrays. Entire borrow, return, and search flow operates on linear data structures.

CONCLUSION :
A practical implementation of linear data structures powering an academic library management system. Arrays, hash maps, and queue patterns are applied across user management, book tracking, and fine calculation workflows.

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